

TV-REMOTE CONTROLLED ROVER WITH OBSTACLE DETECTION

Mini Project Report

Submitted by

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*In partial fulfillment of the requirements for award of the degree of **Bachelor of Technology** in Electronics & Communication Engineering*



Focus on Excellence

Department of Electronics & Communication Engineering

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CERTIFICATE

This is to certify that the mini project report titled **TV-Remote Controlled Rover With Obstacle Detection** submitted by **Johaam J.J, Jyothis George Thaliath, Sarath N.S and Shyamprasad M.P**, towards partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in Electronics and Communication Engineering is a record of bonafide work carried out by them during the academic year 2010 –2011

Staff in charge

Head of the Department

Place:

Date:

Internal Examiner:

External Examiner

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Above all, we owe our heartfelt gratitude to God almighty for all the blessings he has showered on us during this endeavor.

ABSTRACT

The project comprises of a two motor-driven vehicle, carrying an IR receiver which is a TSOP module. This receiver module captures coded IR signals transmitted from an IR TV remote. These coded signals from the remote are decoded using the program contained in the microcontroller and the key being pressed on the TV remote is determined. The microcontroller then issues appropriate commands to the motor driver circuit to move the vehicle in the desired direction.

The motor driver circuit determines the direction of rotation of either of the two wheels. Different directions of motion are obtained by different combinations of wheel movements like: forward, backward, clockwise, and counter-clockwise. Whenever the button on the remote is released, then the vehicle places itself in an idle state.

The vehicle carries proximity sensors on the front and sides. These are comprised of IR LED and photodiodes. Whenever an obstacle comes up in the path of the vehicle, the light from the IR LED gets bounced back from the obstacle and is captured by the photodiode. The microcontroller probes the photodiode to detect any obstructions in the path of the vehicle. Thus the microcontroller is able to avert these obstructions and avoid collisions. They are designed for variable sensitivity depending on the application.

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1. Introduction

1. INTRODUCTION

The project was developed keeping in mind the necessity of controlling and directing devices from a distance. Though there are many different techniques for communicating with a distant device, Infrared signaling seems the most efficient and reliable medium of communication for short to medium range of distances.

The project comprises of a two motor-driven vehicle, carrying an IR receiver which is a TSOP module. This receiver module captures coded IR signals transmitted from an IR TV remote. These coded signals from the remote are decoded using the program contained in the microcontroller and the key being pressed on the TV remote is determined. The microcontroller then issues appropriate commands to the motor driver circuit to move the vehicle in the desired direction.

The motor driver circuit determines the direction of rotation of either of the two wheels. Different directions of motion are obtained by different combinations of wheel movements like: forward, backward, clockwise, and counter-clockwise. Whenever the button on the remote is released, then the vehicle places itself in an idle state.

Even though controlled by a human being, the vehicle could hit obstacles in its path due to human negligence. This becomes a major problem in normal road-vehicles as accidents happen mostly due to human negligence. So there must be some sort of obstacle detection, avoidance and warning system onboard the vehicle to prevent such collisions.

For the very same purpose, the vehicle carries proximity sensors on the front and sides. These are comprised of IR LED and photodiodes. Whenever an obstacle comes up in the path of the vehicle, the light from the IR LED gets bounced back from the obstacle and is captured by the photodiode. The microcontroller probes the photodiode to detect any obstructions in the path of the vehicle. Thus the microcontroller is able to avert these obstructions and avoid collisions. They are designed for variable sensitivity depending on the application. They also provide automatic compensation for measurement errors due to stray light sources like sun or light bulbs, etc.

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2. Component Description

2. COMPONENT DESCRIPTION

2.1 Atmel ATMEGA328P Microprocessor:

The high-performance Atmel picoPower 8-bit AVR RISC-based microcontroller combines 16KB ISP flash memory with read-while-write capabilities, 512B EEPROM, 1KB SRAM, 23 general purpose I/O lines, 32 general purpose working registers, three flexible timer/counters with compare modes, internal and external interrupts, serial programmable USART, a byte-oriented 2-wire serial interface, SPI serial port, a 6-channel 10-bit A/D converter (8-channels in TQFP and QFN/MLF packages), programmable watchdog timer with internal oscillator, and five software selectable power saving modes. The device operates between 2.7-5.5 volts. By executing powerful instructions in a single clock cycle, the device achieves throughputs approaching 1 MIPS per MHz, balancing power consumption and processing speed.

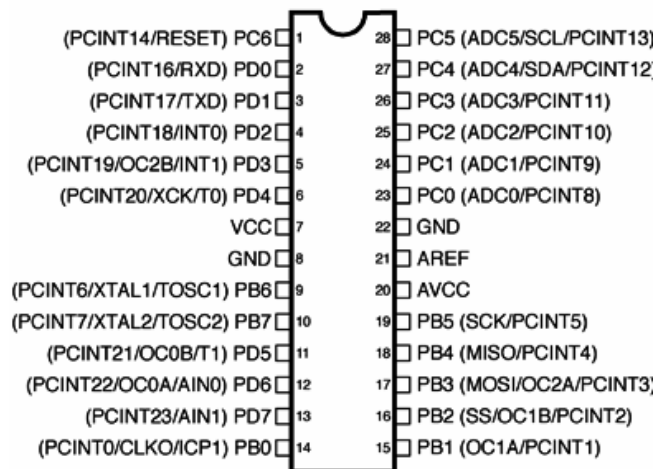


Fig 2.1 Pinout diagram of Atmel ATMEGA 328P

2.2 IR-LED and Photodiode :

IR Diode is an LED which emits radiations in the infrared region. Infrared radiations are invisible. It is used in various IR communication devices. Photodiode generates a voltage when irradiated with light. The one used here is designed to respond to radiations in the infrared region.

2.3 L293D H-Bridge :

The L293 and L293D are quadruple high-current half-H drivers. The L293 is designed to provide bidirectional drive currents of up to 1 A at voltages from 4.5 V to 36 V. The L293D is designed to provide bidirectional drive currents of up to 600-mA at voltages from 4.5 V to 36 V. Both devices are designed to drive inductive loads such as relays, solenoids, dc and bipolar stepping motors, as well as other high-current/high-voltage loads in positive-supply applications. When an enable input is high, the associated drivers are enabled and their outputs are active and in phase with their inputs. When the enable input is low, those drivers are disabled and their outputs are off and in the high-impedance state.

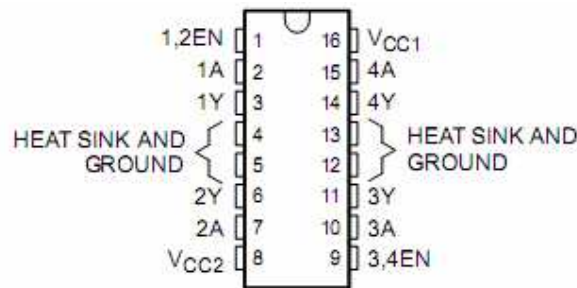


Fig 2.2 Pinout diagram of IC L293D

2.4 Geared DC Motor :

This motor is used to drive the vehicle. Two of these are placed on either side of the vehicle and are connected to wheels. The direction of rotation depends on the polarity of the voltage applied across its terminals. Geared motors are used for low speed applications which require high torque and load-bearing capacity.

2.5 TSOP 1738 IR Module :

The TSOP1738 is a miniaturized receiver for infrared remote control systems. It consists of a PIN photodiode and a preamplifier stage enclosed in an epoxy case. Its output is active low and gives +5 V when off. The demodulated output can be directly decoded by a microprocessor. The important features of the module includes internal filter for PCM frequency, TTL and CMOS compatibility, low power consumption (5

volt and 5 mA), immunity against ambient light, noise protection etc. The added features are continuous data transmission up to 2400 bps and suitable burst length of 10 cycles per burst.

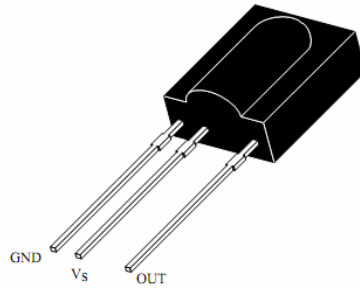


Fig 2.3 Pinout diagram of TSOP 1738

2.6 LM7805 Voltage Regulator :

The 7805 IC is a three-terminal positive voltage regulator available in the TO-220/D-PAK package. Typically provide 1 or 1.5 amps of current (though smaller or larger packages may have a lower or higher current rating). Although designed primarily as fixed voltage regulators, these devices can be used with external components to obtain adjustable voltages and currents.

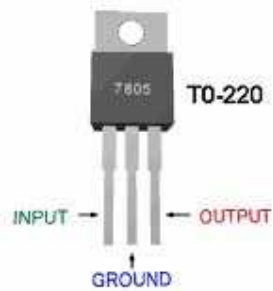


Fig. 2.4 Pinout diagram of LM 7805

3. Block Diagram

3. BLOCK DIAGRAM

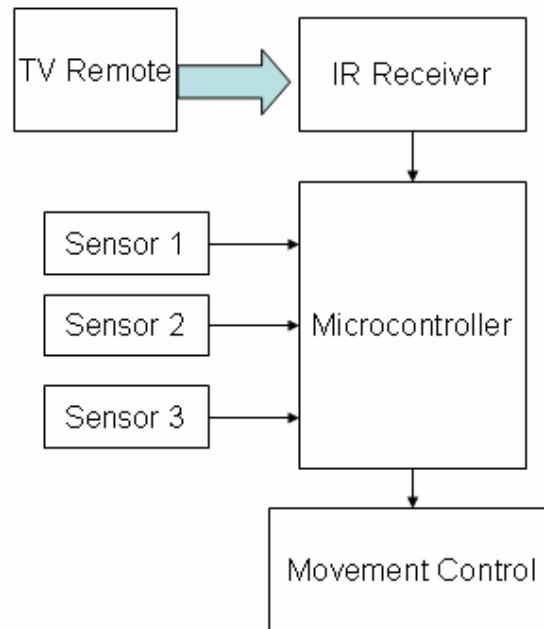


Fig. 3.1: Block Diagram

The main parts of the rover are as shown in these blocks. It consists of a microcontroller which controls all the processing part of the system. It decodes the IR signals into meaningful commands and gives directions to the motor driver circuit. The IR receiver is a module that captures IR pulses of a particular frequency which is coming from a TV-remote. The motor driver circuit is a group of switches which determine the supply and polarity of the power input to the driving motors. Its action depends on the control signals issued by the microcontroller. The sensors form three different modules place on front, and both sides of the rover. These help avoid collisions by detecting any forms of obstructions in the path of the moving vehicle.

4. Circuit Diagram

4. CIRCUIT DIAGRAM

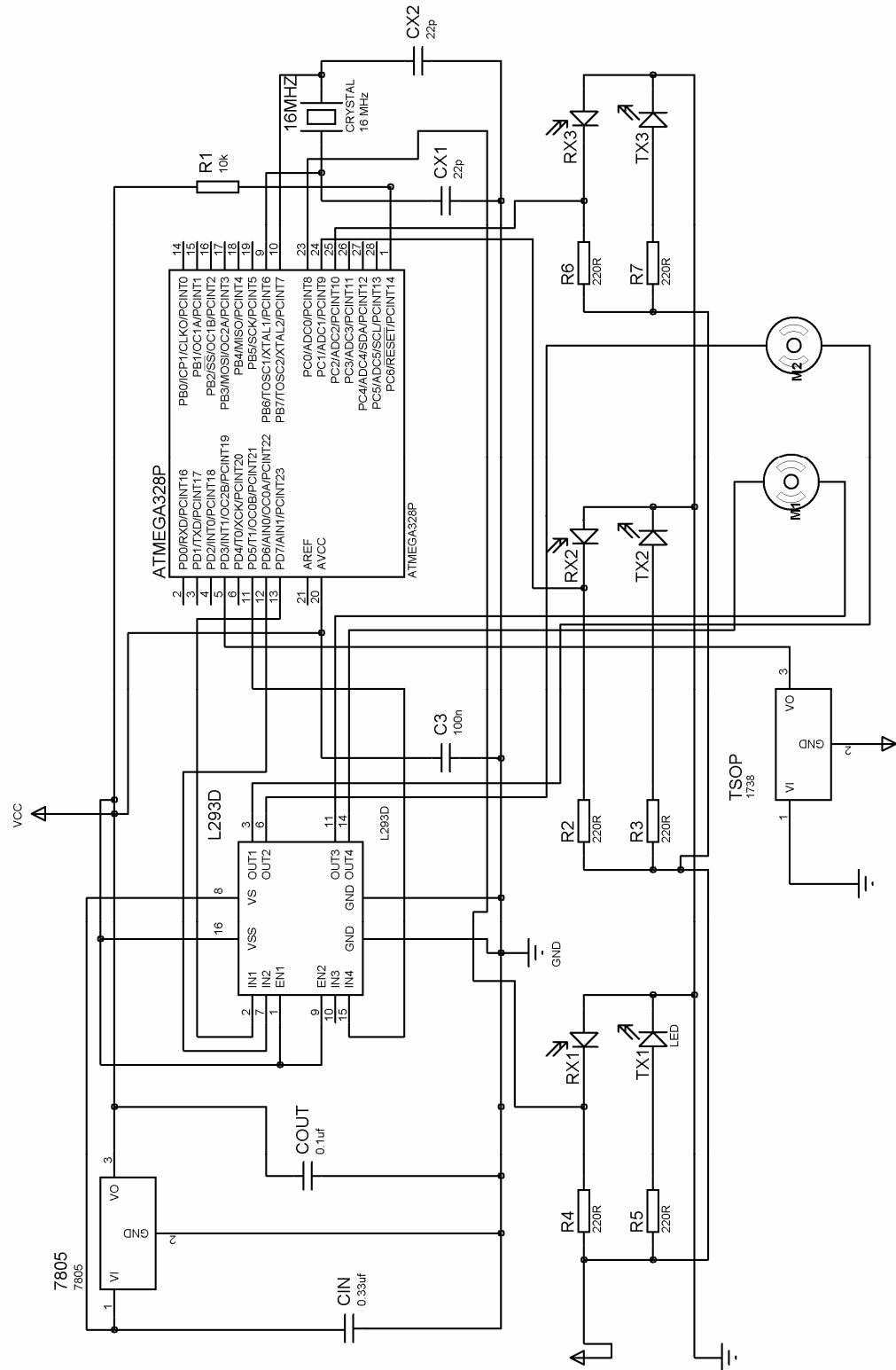


Fig. 4.1: Circuit Diagram

5. Software Section

5. SOFTWARE SECTION

The programming software used is Arduino Alpha available freely from the website www.arduino.cc. The programming language used is a modified form of Wiring C, exclusively used for program development for Arduino boards. Arduino hardware is programmed using a Wiring-based language (syntax + libraries), similar to C++ with some simplifications and modifications, and a Processing-based IDE.

The Arduino IDE comes with a C/C++ library called "Wiring" (from the project of the same name), which makes many common input/output operations much easier. Arduino programs are written in C/C++. The code would not be seen by a standard C++ compiler as a valid program, so when the user clicks the "Upload to I/O board" button in the IDE, a copy of the code is written to a temporary file with an extra include header at the top and a very simple main() function at the bottom, to make it a valid C++ program. The Arduino IDE uses the GNU Toolchain and AVR Libc to compile programs, and uses AVRDUDE to upload programs to the board.

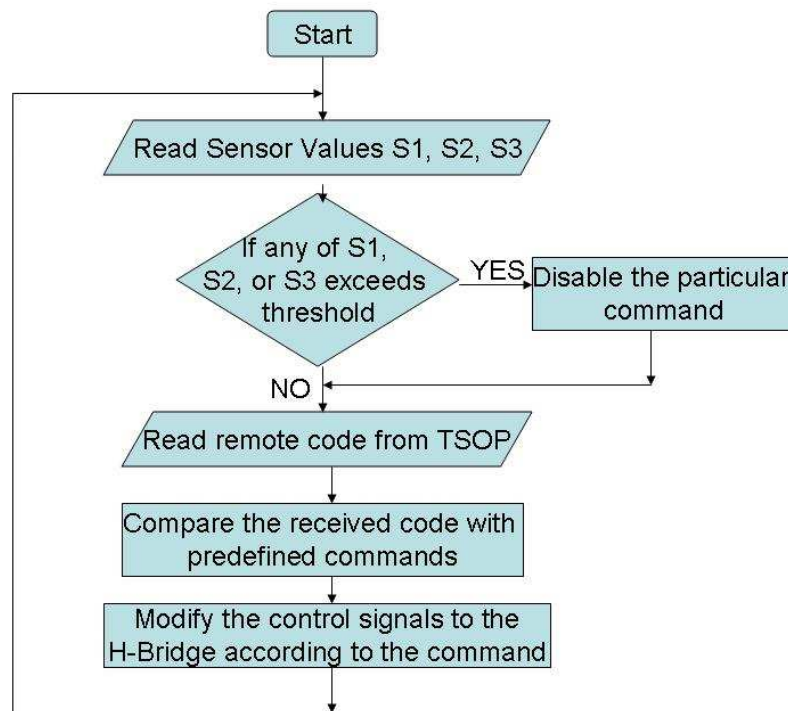


Fig. 5.1: Flowchart

6. IR Transmission Protocol

6. IR TRANSMISSION PROTOCOL

6.1 General format :

The infrared remote control signal starts with a leader code. Next comes a 16-bit custom code, then an 8-bit data code and an inverted binary 8-bit code, and finally a stop bit.

An example of the infrared remote control format is shown below. This signal is followed by a frame space during which no infrared rays are emitted. The total frame length (including everything from the leader to the frame space) is 108 ms.

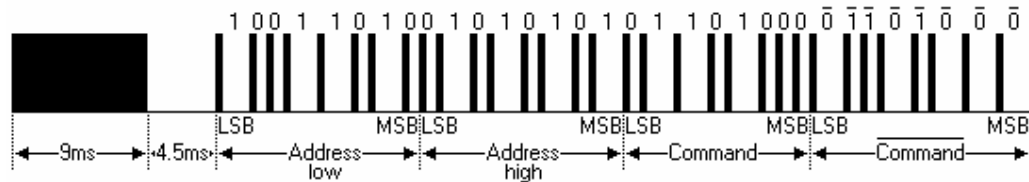


Fig 6.1: Example of NEC format for infrared remote control

6.2 Leader code and repeat code:

The leader code stays ON for a 9-ms period, then is OFF for a 4.5-ms period. Since this part's waveform (timing) differs greatly from the following data code section, it makes the leader code easier to recognize. (When repeating, the OFF period is only 2.25 ms, and the stop bit comes next, omitting the custom code and data codes).

A command is transmitted only once, even when the key on the remote control remains pressed. Every 110ms a repeat code is transmitted for as long as the key remains down. This repeat code is simply a 9ms AGC pulse followed by a 2.25ms space and a 560μs burst.

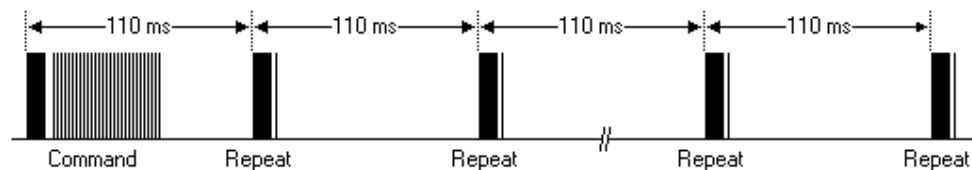


Fig 6.2: Repeat code transmission

6.3 Data transmission sequence :

The structure of remote control signal transmitted via this method consists of custom code and data code. The custom code, which is transmitted first, is 16 bits long but it is divided into two 8-bit sections. In early versions of remote control devices, the custom code was only 8 bits long (C0 to C7), and the logically inverted data (C'0 to C'7) was transmitted via the next 8 bits. Now this C'0 to C'7 section has been reassigned as the second section of the custom code so that the custom code is 16 bits long. (16-bit data is specified as the sum of custom code = $xx + \text{custom code}' = yy$.) When transmitting, the custom code is output LSB first (C0 to C7), then the custom code' is output LSB first (C'0 to C'7).

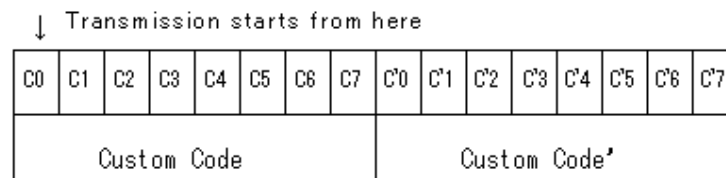


Fig 6.3 Transmission sequence of custom code section

The data being transmitted is 8-bit data. The logically inverted 8-bit data is transmitted continuously, so a total of 16 bits are used to transmit the data. When this data is received, the inverted 8-bit data code should be checked as being the logical inversion of the first 8-bit data code, as a means of error checking.

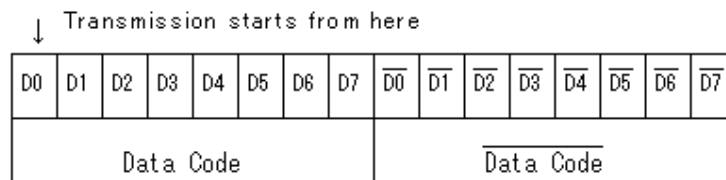


Fig 6.4 Transmission sequence of data code sections

7. PCB Fabrication

7. PCB FABRICATION

7.1 Layout preparation :

1. Each and every PCB layout is viewed from component side.
2. The larger components are placed first and space between them is filled.
3. In the designing of the PCB layout, it is very important to divide the circuit into functional subunits, each of these subunits in the defined portion of the boards.
4. The components are placed in the grid sheet tanning the standard length and width.
5. The punched component layout is circled to take the standard size of the land pads.
6. These parts are connected as per the circuit diagram.
7. The mirror images of these gives the solder side of the PCB.

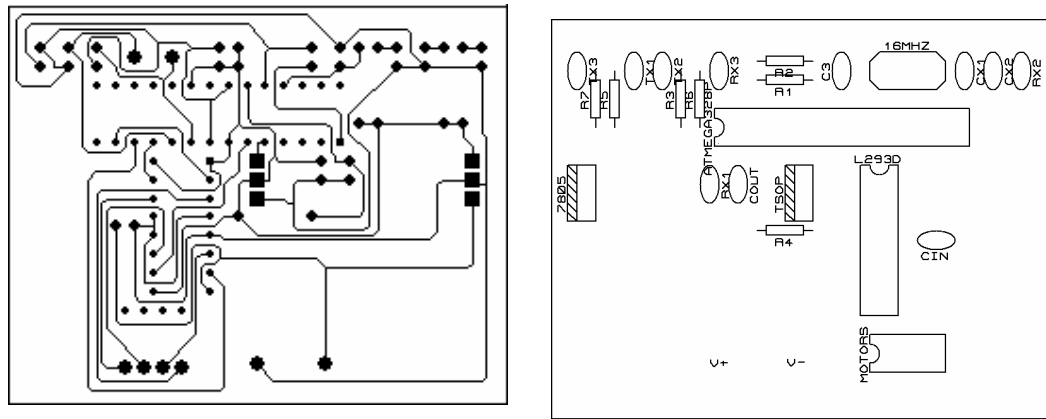


Fig 7.1 PCB Layout and Component Layout

7.2 Screen printing :

In screen printing, the process is very simple. A screen fabric with uniform meshes and opening is stretched and fixed on a solid frame of metal or wood. The circuit pattern is photographically transferred on the screen, leaving the meshes in the rest of the area as closed. In the actual printing step is forced by moving queue through the open master on to the surface of the material to be printed. The light sensitive material is coated on to the screen and using the film master, the pattern is transferred on the screen. Then, using ink the pattern is transferred to the copper clad sheet.

7.3 Etching :

The removal of unwanted copper from the copper clad sheet is known as etching. For this four types of tanks are used.

1. Ferric chloride
2. Cupric chloride.
3. Chromic acid.
4. Alkaline ammonia.

Among this ferric chloride is cheap and more popular etch and it is also suited for home and industrial applications.

7.4 Drilling :

Drilling of components by mounting by mounting holes into PCBs is the most important mechanical machining operation for PCB industrial production process. The importance of hole drilling on PCB has further group with electronic component miniaturization and its need for smaller hole diameters and higher packages density where hole punching is practically routed out.

7.5 Component mounting :

Components are basically mounted on one side of the board. Polarized two-lead components are mounted to give the marking of the orientation throughout the board. The component orientation can be both horizontal and vertical. The uniformity in orientation of components can be determined during the design of PCB.

7.6 Soldering :

Soldering is an important process in assembling electronic products. Solder joints are formed by nature of welding processes. Solder does not stick on the insulating surface. On most of the metals welding will take place and a joint will be formed if the work piece and solder are hot enough and the surfaces are clean and free from oxides.

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8. Results

8. RESULTS

The circuit was developed in various stages. Individual stages were wired and tested using prototype board and Arduino UNO development board. After testing individual stages, the whole circuit was wired and tested on prototype board. Its working was verified after sufficient testing.

After completion of prototyping, the circuit had to be transferred into a PCB. Layout was created using freely available software like: Eagle and Proteus. This layout was then developed into a PCB after processes of printing, etching etc. various components were soldered onto the board according to the given layout.

The circuit was once again tested fully using the developed PCB. Certain changes were applied to the initial program to compensate for the changes occurred while moving from prototype to PCB. It was noted that the power requirement has increased considerably after moving on to PCB. Certain power drops were experienced like the motors slowing down. So, batteries could no longer be used, but it requires an external DC source to power the circuit. This drastically reduced the degree of freedom of the vehicle.

The circuit is found to be fully complying with its indented use. The rover is responding remarkably well to the remote control. The estimated range of the controller is about 10 meters from the rover. Though the speed of the vehicle was reduced, it infact turned out to be a boon as the sensing time of the obstacle detectors is far higher when the vehicle is moving slowly. So it can respond more quickly to avoid collisions. The obstacle sensors were designed to be of variable sensitivity. They can be set to a threshold distance level from 4cms upto 20cms depending on the indented purpose.

9. Conclusion and Future Scope

9. CONCLUSION AND FUTURE SCOPE

The project turned out to be successful. A thorough study has been made in understanding the concepts of IR based communication techniques. It is possible to replace it with any other form of digital communication techniques like GSM, Bluetooth, etc. Only the principles of transmission vary from technique to technique. Also, it led to a good exposure to the field of robotics. It made awareness about different possibilities of controlling the movements and obtaining desired results using microcontrollers. The same techniques can be adapted into robots of any form and size to be controlled in a similar fashion. Only the control circuits need modifications.

Members of the group put forward a suggestion to improve the existing system by adding an additional feature, that is, an Automated Mode for fully automatic control of the rover by the microprocessor alone without human intervention. The proposal was welcoming as it opened up a new and exciting field of Artificial Intelligence into our project. The hardware implementation was not at all difficult because the rover already has an obstacle avoidance system. Only changes required is in the software part.

The application of such a feature is evident in the modern world where road cars are becoming increasingly intelligent. One day they would be capable of driving themselves without human intervention. Such a system would not only reduce the human effort in driving but would prove drastic changes in the ways accidents occur due to human negligence.

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APPENDIX

Program :

```
#include <IRremote.h>

bool m1a,m2a,m1b,m2b,mod=LOW,pwr=HIGH,
fwd_dis=LOW,left_dis=LOW,right_dis=LOW;
int x=0,y=0,amb0=0,amb1=0,amb2=0,M1A=7,M1B=6,M2A=5,M2B=4,MOD=8,
RCV_PIN = 3,PWR_LED=13, s1=0, s2=0, s0=0;

static int j;

IRrecv irrecv(RCV_PIN);

decode_results results;

void setup()
{
    Serial.begin(9600);
    irrecv.enableIRIn(); // Start the receiver
    pinMode(M1A, OUTPUT);    pinMode(M1B, OUTPUT);    pinMode(M2A,
OUTPUT);
    pinMode(M2B, OUTPUT);    pinMode(PWR_LED, OUTPUT);
    amb0 = analogRead(0);    amb1 = analogRead(1);    amb2 = analogRead(2);
}

int signal()
{
    int i;
    if (irrecv.decode(&results))
    {
        switch(results.value)
        {
            case 0xFF52AD : i= 1; break;
            case 0xFF7887 : i= 2; break;
            case 0xFF42BD : i= 3; break;
            case 0xFFB847 : i= 4; break;
            case 0xFFE21D : i= 5; break;
            case 0xFF02FD : i=6; break;
            case 0xFFFFFFFF : i= j; break;
        }
    }
}
```

```

    irrecv.resume(); // Receive the next value
    j=i;
}
else i=0;
return i;
}
void fwd()
{
    if(fwd_dis==LOW)
        m1a=HIGH; m1b=LOW; m2a=HIGH; m2b=LOW;
}
void back()
{
    m1a=LOW; m1b=HIGH; m2a=LOW; m2b=HIGH;
}
void left()
{
    if(left_dis==LOW)
    {
        m1a=LOW; m1b=HIGH; m2a=HIGH; m2b=LOW;
    }
}
void right()
{
    if(right_dis==LOW)
    {
        m1a=HIGH; m1b=LOW; m2a=LOW; m2b=HIGH;
    }
}
void idle()
{
    m1a=LOW; m1b=LOW; m2a=LOW; m2b=LOW; }
void mode()
{
    if (mod==LOW)

```



```

        mod=HIGH;
    else
        mod=LOW;
    }
void power()
{
    if (pwr==LOW)
    {
        pwr=HIGH;
        mod=LOW;
    }
    else
        pwr=LOW;
}
void drive()
{
    digitalWrite(M1A,m1a);    digitalWrite(M2A,m2a);
    digitalWrite(M1B,m1b);    digitalWrite(M2B,m2b);    digitalWrite(MOD,mod);
    delay(100);
    idle();
}
void sense()
{
    s0 = analogRead(0); s1 = analogRead(1);  s2 = analogRead(2);
    if(s0>(amb0+80))
        fwd_dis=HIGH;
    else
        fwd_dis=LOW;
    if(s1>(amb1+20))
        left_dis=HIGH;

    else
        left_dis=LOW;
    if(s2>(amb2+20))
        right_dis=HIGH;

```

```

else
    right_dis=LOW;
}
void loop()
{
    digitalWrite(PWR_LED,pwr);
    sense();
    x=signal();
    if(x!=0||(x==0&&y==0))
    {
        if(x!=5 && mod==LOW && pwr==HIGH) //REMOTE DRIVE
        {
            switch(x)
            {
                case 0 :      idle();      break;
                case 1 :      fwd();       break;
                case 2 :      back();      break;
                case 3 :      left();      break;
                case 4 :      right();     break;
            }
            drive();
        }
        else if(x==5 && pwr==HIGH) //AUTO DRIVE
            mode();
        if(x==6) //POWER CHANGE
            power();
    }
    y=x;
}

```

Features

- High Performance, Low Power AVR[®] 8-Bit Microcontroller
- Advanced RISC Architecture
 - 131 Powerful Instructions – Most Single Clock Cycle Execution
 - 32 x 8 General Purpose Working Registers
 - Fully Static Operation
 - Up to 20 MIPS Throughput at 20 MHz
 - On-chip 2-cycle Multiplier
- High Endurance Non-volatile Memory Segments
 - 4/8/16/32K Bytes of In-System Self-Programmable Flash program memory
 - 256/512/1K Bytes EEPROM
 - 512/1K/1K/2K Bytes Internal SRAM
 - Write/Erase Cycles: 10,000 Flash/100,000 EEPROM
 - Data retention: 20 years at 85°C/100 years at 25°C⁽¹⁾
 - Optional Boot Code Section with Independent Lock Bits
 - In-System Programming by On-chip Boot Program
 - True Read-While-Write Operation
 - Programming Lock for Software Security
- Peripheral Features
 - Two 8-bit Timer/Counters with Separate Prescaler and Compare Mode
 - One 16-bit Timer/Counter with Separate Prescaler, Compare Mode, and Capture Mode
 - Real Time Counter with Separate Oscillator
 - Six PWM Channels
 - 8-channel 10-bit ADC in TQFP and QFN/MLF package
 - Temperature Measurement
 - 6-channel 10-bit ADC in PDIP Package
 - Temperature Measurement
 - Programmable Serial USART
 - Master/Slave SPI Serial Interface
 - Byte-oriented 2-wire Serial Interface (Philips I²C compatible)
 - Programmable Watchdog Timer with Separate On-chip Oscillator
 - On-chip Analog Comparator
 - Interrupt and Wake-up on Pin Change
- Special Microcontroller Features
 - Power-on Reset and Programmable Brown-out Detection
 - Internal Calibrated Oscillator
 - External and Internal Interrupt Sources
 - Six Sleep Modes: Idle, ADC Noise Reduction, Power-save, Power-down, Standby, and Extended Standby
- I/O and Packages
 - 23 Programmable I/O Lines
 - 28-pin PDIP, 32-lead TQFP, 28-pad QFN/MLF and 32-pad QFN/MLF
- Operating Voltage:
 - 1.8 - 5.5V
- Temperature Range:
 - -40°C to 85°C
- Speed Grade:
 - 0 - 4 MHz@1.8 - 5.5V, 0 - 10 MHz@2.7 - 5.5V, 0 - 20 MHz @ 4.5 - 5.5V
- Power Consumption at 1 MHz, 1.8V, 25°C
 - Active Mode: 0.2 mA
 - Power-down Mode: 0.1 µA
 - Power-save Mode: 0.75 µA (Including 32 kHz RTC)



8-bit AVR[®]
Microcontroller
with 4/8/16/32K
Bytes In-System
Programmable
Flash

ATmega48A
ATmega48PA
ATmega88A
ATmega88PA
ATmega168A
ATmega168PA
ATmega328
ATmega328P

Summary

Rev. 8271CS-AVR-08/10



1.1 Pin Descriptions

1.1.1 VCC

Digital supply voltage.

1.1.2 GND

Ground.

1.1.3 Port B (PB7:0) XTAL1/XTAL2/TOSC1/TOSC2

Port B is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The Port B output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port B pins that are externally pulled low will source current if the pull-up resistors are activated. The Port B pins are tri-stated when a reset condition becomes active, even if the clock is not running.

Depending on the clock selection fuse settings, PB6 can be used as input to the inverting Oscillator amplifier and input to the internal clock operating circuit.

Depending on the clock selection fuse settings, PB7 can be used as output from the inverting Oscillator amplifier.

If the Internal Calibrated RC Oscillator is used as chip clock source, PB7...6 is used as TOSC2...1 input for the Asynchronous Timer/Counter2 if the AS2 bit in ASSR is set.

The various special features of Port B are elaborated in [and "System Clock and Clock Options" on page 26](#).

1.1.4 Port C (PC5:0)

Port C is a 7-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The PC5...0 output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port C pins that are externally pulled low will source current if the pull-up resistors are activated. The Port C pins are tri-stated when a reset condition becomes active, even if the clock is not running.

1.1.5 PC6/ $\overline{\text{RESET}}$

If the RSTDISBL Fuse is programmed, PC6 is used as an I/O pin. Note that the electrical characteristics of PC6 differ from those of the other pins of Port C.

If the RSTDISBL Fuse is unprogrammed, PC6 is used as a Reset input. A low level on this pin for longer than the minimum pulse length will generate a Reset, even if the clock is not running. The minimum pulse length is given in [Table 28-12 on page 323](#). Shorter pulses are not guaranteed to generate a Reset.

The various special features of Port C are elaborated in ["Alternate Functions of Port C" on page 86](#).

1.1.6 Port D (PD7:0)

Port D is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The Port D output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port D pins that are externally pulled low will source current if the pull-up resistors are activated. The Port D pins are tri-stated when a reset condition becomes active, even if the clock is not running.

The various special features of Port D are elaborated in ["Alternate Functions of Port D" on page 89](#).

1.1.7 **AV_{CC}**

AV_{CC} is the supply voltage pin for the A/D Converter, PC3:0, and ADC7:6. It should be externally connected to V_{CC}, even if the ADC is not used. If the ADC is used, it should be connected to V_{CC} through a low-pass filter. Note that PC6...4 use digital supply voltage, V_{CC}.

1.1.8 **AREF**

AREF is the analog reference pin for the A/D Converter.

1.1.9 **ADC7:6 (TQFP and QFN/MLF Package Only)**

In the TQFP and QFN/MLF package, ADC7:6 serve as analog inputs to the A/D converter. These pins are powered from the analog supply and serve as 10-bit ADC channels.

KA78XX/KA78XXA

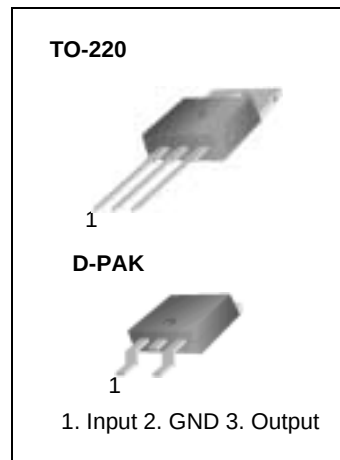
3-Terminal 1A Positive Voltage Regulator

Features

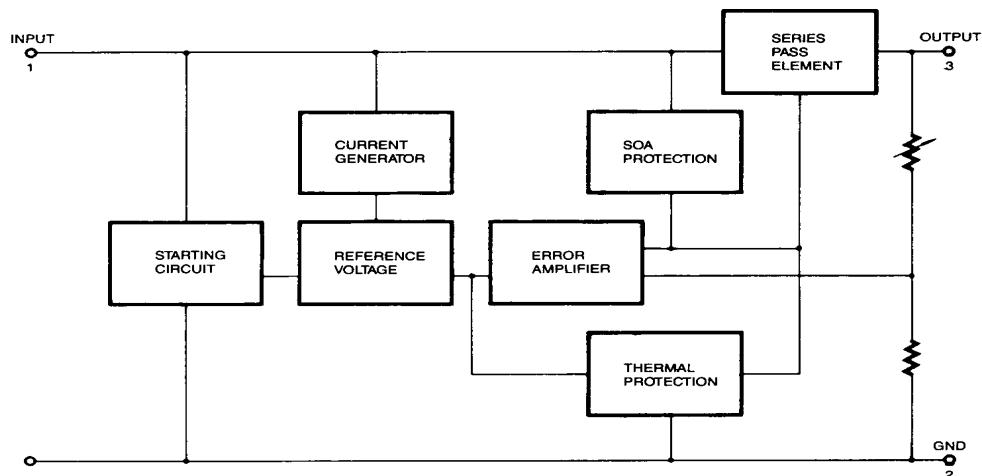
- Output Current up to 1A
- Output Voltages of 5, 6, 8, 9, 10, 12, 15, 18, 24V
- Thermal Overload Protection
- Short Circuit Protection
- Output Transistor Safe Operating Area Protection

Description

The KA78XX/KA78XXA series of three-terminal positive regulator are available in the TO-220/D-PAK package and with several fixed output voltages, making them useful in a wide range of applications. Each type employs internal current limiting, thermal shut down and safe operating area protection, making it essentially indestructible. If adequate heat sinking is provided, they can deliver over 1A output current. Although designed primarily as fixed voltage regulators, these devices can be used with external components to obtain adjustable voltages and currents.



Internal Block Diagram



L293, L293D QUADRUPLE HALF-H DRIVERS

SLRS008B – SEPTEMBER 1986 – REVISED JUNE 2002

- Featuring Unitrode L293 and L293D Products Now From Texas Instruments
- Wide Supply-Voltage Range: 4.5 V to 36 V
- Separate Input-Logic Supply
- Internal ESD Protection
- Thermal Shutdown
- High-Noise-Immunity Inputs
- Functional Replacements for SGS L293 and SGS L293D
- Output Current 1 A Per Channel (600 mA for L293D)
- Peak Output Current 2 A Per Channel (1.2 A for L293D)
- Output Clamp Diodes for Inductive Transient Suppression (L293D)

description

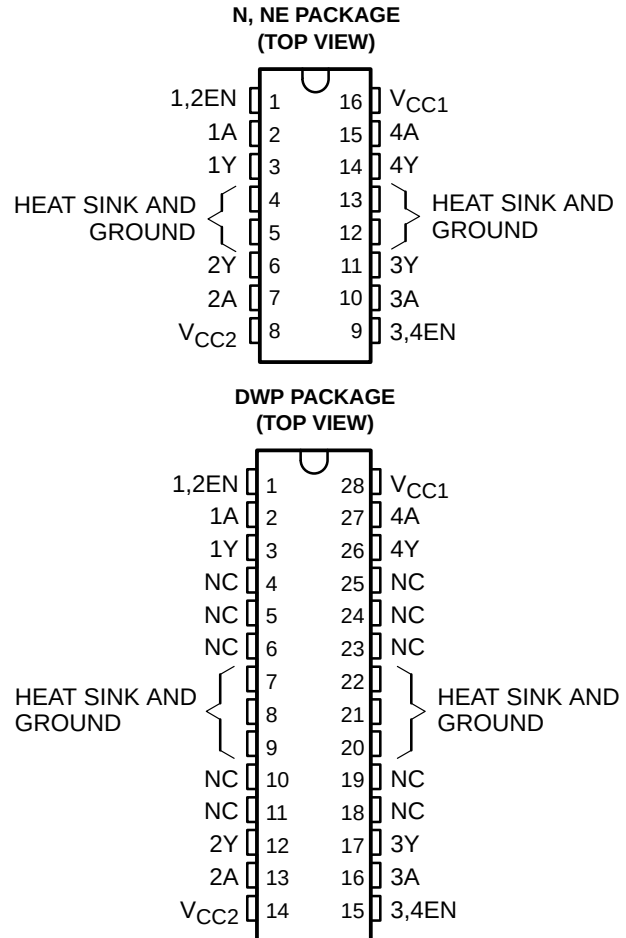
The L293 and L293D are quadruple high-current half-H drivers. The L293 is designed to provide bidirectional drive currents of up to 1 A at voltages from 4.5 V to 36 V. The L293D is designed to provide bidirectional drive currents of up to 600-mA at voltages from 4.5 V to 36 V. Both devices are designed to drive inductive loads such as relays, solenoids, dc and bipolar stepping motors, as well as other high-current/high-voltage loads in positive-supply applications.

All inputs are TTL compatible. Each output is a complete totem-pole drive circuit, with a Darlington transistor sink and a pseudo-Darlington source. Drivers are enabled in pairs, with drivers 1 and 2 enabled by 1,2EN and drivers 3 and 4 enabled by 3,4EN. When an enable input is high, the associated drivers are enabled and their outputs are active and in phase with their inputs. When the enable input is low, those drivers are disabled and their outputs are off and in the high-impedance state. With the proper data inputs, each pair of drivers forms a full-H (or bridge) reversible drive suitable for solenoid or motor applications.

On the L293, external high-speed output clamp diodes should be used for inductive transient suppression.

A V_{CC1} terminal, separate from V_{CC2} , is provided for the logic inputs to minimize device power dissipation.

The L293 and L293D are characterized for operation from 0°C to 70°C.



Please be aware that an important notice concerning availability, standard warranty, and use in critical applications of Texas Instruments semiconductor products and disclaimers thereto appears at the end of this data sheet.

PRODUCTION DATA information is current as of publication date. Products conform to specifications per the terms of Texas Instruments standard warranty. Production processing does not necessarily include testing of all parameters.

**TEXAS
INSTRUMENTS**

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Photo Modules for PCM Remote Control Systems

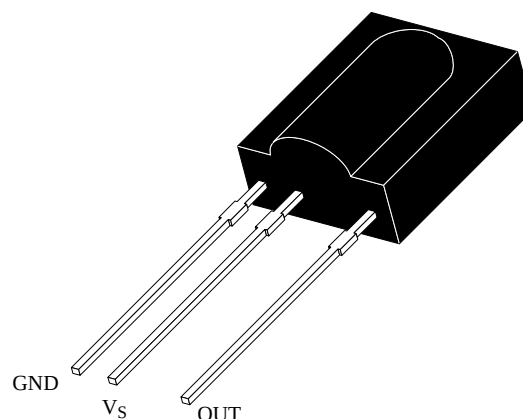
Available types for different carrier frequencies

Type	fo	Type	fo
TSOP1730	30 kHz	TSOP1733	33 kHz
TSOP1736	36 kHz	TSOP1737	36.7 kHz
TSOP1738	38 kHz	TSOP1740	40 kHz
TSOP1756	56 kHz		

Description

The TSOP17.. – series are miniaturized receivers for infrared remote control systems. PIN diode and preamplifier are assembled on lead frame, the epoxy package is designed as IR filter.

The demodulated output signal can directly be decoded by a microprocessor. TSOP17.. is the standard IR remote control receiver series, supporting all major transmission codes.

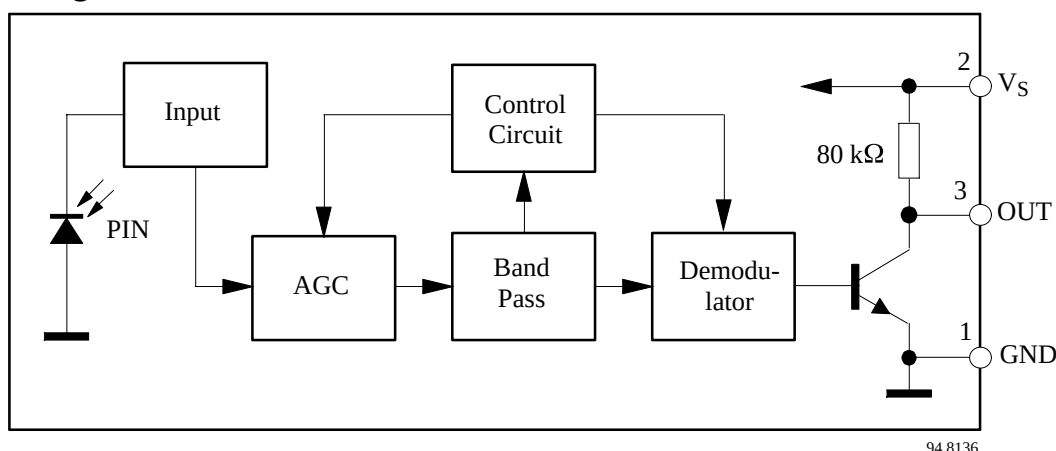


94 8691

Features

- Photo detector and preamplifier in one package
- Internal filter for PCM frequency
- Improved shielding against electrical field disturbance
- TTL and CMOS compatibility
- Output active low
- Low power consumption
- High immunity against ambient light
- Continuous data transmission possible (up to 2400 bps)
- Suitable burst length ≥ 10 cycles/burst

Block Diagram



94 8136